Worked Calculations

EEE3094S Lab 2 – PTRJUN003

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1 Model, Inputs and Definitions

The original CSV and screenshot are retained in the data folder. The Python modules calculate the values below; MATLAB design code provides the root-locus method. This supplement contains worked substitutions rather than additional experiments.

$$\Delta u = 5.0 - 2.5 = 2.5\,\mathrm{V}, \qquad u_h = 2.5\,\mathrm{V}, \tag{1.1}$$

$$G_y(s) = \frac{14.2748}{s(8.1s+1)}, \qquad k_s = 0.93\,\mathrm{V/m}, \tag{1.2}$$

$$b = \frac{1}{8.1} = 0.123456790\,\mathrm{s}^{-1}, \qquad k_T = \frac{14.2748}{8.1} = 1.762320988, \tag{1.3}$$

$$L(s) = K_p k_s G_y(s), \qquad K_p = 0.01432. \tag{1.4}$$

The closed-loop reference is a position step from 0 to 5 m; the identification input is a voltage step from 2.5 to 5.0 V. They are different signals.

$$k_s = \frac{\sum_i y_i y_{v,i}}{\sum_i y_i^2}. \tag{1.5}$$

$$M_p = 100e^{-\pi\zeta/\sqrt{1-\zeta^2}}. \tag{1.6}$$

$$y(t) = 5\left[1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right)\right]. \tag{1.7}$$

$$\delta y = \frac{\delta u_b}{K_p k_s}. \tag{1.8}$$

The circuit coefficients used for tolerance calculations are

$$a_f = R_4/R_3, \qquad a_r = (1+a_f)R_2/(R_1+R_2), \tag{1.9}$$

$$g_2 = 1 + R_5/R_6, \qquad g_3 = (1 + R_{10}/(R_9 + R_{\mathrm{th}}))R_8/(R_7+R_8). \tag{1.10}$$

2 Detailed Calculations

2.1 Model Derivation and Signal Definitions

Table 2.1: Model notation and units.

Symbol	Description	Unit
$x, y, v = \dot{y}$	Reference altitude, altitude, vertical velocity	m, m, m/s
f_t, g, d_a	Mass-normalised thrust, gravity, disturbance	$\mathrm{m/s^2}$
b, k_T	Drag coefficient; local voltage-to-thrust slope	$\mathrm{s^{-1}}$; $\mathrm{m/(s^2\ V)}$
$u, u_h, \delta u$	Command, hover command, $u - u_h$	V
A, τ	Velocity gain; velocity time constant	m/(s V); s
k_s	Position-to-voltage coefficient	V/m
r_v, y_v, e_v	Reference, sensor, error voltage	V
K_p	Proportional voltage-error gain	V/V

Start from $\ddot{y} = f_t - b\dot{y} - g + d_a$, with $f_t = g + k_T(u - u_h)$. For $d_a = 0$, zero initial conditions and $v = \dot{y}$,

$$\dot{v} + bv = k_T \delta u, \tag{2.1}$$

$$(s + b)V(s) = k_T \Delta U(s), \tag{2.2}$$

$$G_v(s) = \frac{k_T}{s + b} = \frac{k_T/b}{s/b + 1} = \frac{A}{\tau s + 1}, \tag{2.3}$$

$$G_y(s) = \frac{G_v(s)}{s} = \frac{A}{s(\tau s + 1)}. \tag{2.4}$$

The identified parameters give $b = 1/8.1 = 0.123456790\ \mathrm{s^{-1}}$ and $k_T = 14.2748/8.1 = 1.762320988\ \mathrm{m/(s^2\ V)}$. The hover voltage does not imply $k_T = g/u_h$.

For negative feedback, $E_y = X - Y$ and $L = K_p k_s G_y$:

$$\frac{E_y(s)}{X(s)} = \frac{1}{1 + L(s)} = \frac{\tau s^2 + s}{\tau s^2 + s + AK_p k_s}. \tag{2.5}$$

For a step $X = x_0/s$, the final-value theorem gives $e_\infty = \lim_{s \to 0} sE_y(s) = 0$. For a ramp $X = a_r/s^2$,

$$e_\infty = \frac{a_r}{AK_p k_s}, \qquad K_v = AK_p k_s = 0.190106076. \tag{2.6}$$

A parabolic reference has unbounded steady error. A constant acceleration disturbance d_0 instead gives $y_\infty - x_\infty = d_0/(k_T K_p k_s)$. For an input bias, $d_0 = k_T \delta u_b$, so $\delta y = \delta u_b/(K_p k_s)$.

2.2 Identification and Timing

Table 2.2 gives selected original samples and derived velocities. The finite manual step is evident in the input column. Endpoint times are listed for consistency with the CSV; each derivative physically belongs to its interval midpoint.

Table 2.2: Selected identification samples and backward-difference results.

t [s]	u [V]	y_v [V]	y [m]	v_i [m/s]
15.7	2.495	0.000	0.000	0.00
15.8	2.908	0.000	0.000	0.00
15.9	4.607	0.000	0.000	0.00
16.0	5.000	0.034	0.037	0.37
16.9	5.000	2.098	2.256	4.08
17.0	5.000	2.513	2.703	4.47
23.9	5.000	10.000	104.823	22.36
24.0	5.000	10.000	107.076	22.53
24.1	5.000	10.000	109.345	22.69

$$v(24.0) = (107.076 - 104.823)/0.1 = 22.53\,\mathrm{m/s}, \tag{2.7}$$

$$v(24.1) = (109.345 - 107.076)/0.1 = 22.69\,\mathrm{m/s}, \tag{2.8}$$

$$\bar{v}_\infty = \frac{1}{404}\sum_{t_i \geq 120} \frac{y_i - y_{i-1}}{0.1} = 35.68747525\,\mathrm{m/s}, \tag{2.9}$$

$$A = (35.68747525 - 0)/2.5 = 14.27499010\,\mathrm{m/(s\,V)}, \tag{2.10}$$

$$v_{63} = 0.632(35.68747525) = 22.55448436\,\mathrm{m/s}, \tag{2.11}$$

$$t_{63} = 24.0 + 0.1\frac{22.55448436 - 22.53}{22.69 - 22.53} = 24.01530272\,\mathrm{s}, \tag{2.12}$$

$$\tau_{63} = 24.01530272 - 15.9 = 8.11530272\,\mathrm{s}. \tag{2.13}$$

The exponential fit minimises $\sum_{17 \leq t_i \leq 60}[v_i - V_\infty(1 - e^{-(t_i - t_0)/\tau})]^2$. Its outputs are $V_\infty = 35.68747059$ m/s, $\tau = 8.10393362$ s and $t_0 = 15.91583783$ s. These two methods corroborate rounding to $\tau = 8.1$ s. They do not create two experimental trials.

The sensor least-squares fit in Equation (1.5) excludes zero and clipped pairs. The check $k_s(2.703) = 0.93(2.703) = 2.51379$ V agrees with the recorded 2.513 V within rounding.

2.3 Root-Locus Geometry and Gain Bound

Step 1 – Form the loop without the variable gain. Include the sensor coefficient explicitly:

$$G_0(s) = k_s G_y(s) = \frac{14.2748(0.93)}{s(8.1s+1)} = \frac{1.638958519}{s(s+0.123456790)}. \tag{2.14}$$

Step 2 – Locate poles and asymptotes. The two poles are $p_1 = 0$ and $p_2 = -0.123456790$; there are no finite zeros. The real-axis locus is between the poles. With $n - m = 2$,

$$\sigma_a = \frac{p_1 + p_2}{2} = -0.061728395, \qquad \theta_\ell = \frac{(2\ell+1)180^\circ}{2} = 90^\circ, 270^\circ. \tag{2.15}$$

Step 3 – Calculate breakaway. From $1 + K_p G_0(s) = 0$,

$$K_p(s) = -\frac{s(8.1s+1)}{14.2748(0.93)}, \tag{2.16}$$

$$\frac{dK_p}{ds} = -\frac{16.2s+1}{13.275564} = 0 \quad \Rightarrow \quad s_b = -\frac{1}{16.2} = -0.061728395, \tag{2.17}$$

$$K_{p,b} = \frac{1}{4(8.1)(14.2748)(0.93)} = 0.002324888. \tag{2.18}$$

Step 4 – Apply the overshoot bound. Solving $0.30 = e^{-\pi\zeta/\sqrt{1-\zeta^2}}$ gives

$$\zeta_{\min} = \frac{-\ln 0.30}{\sqrt{\pi^2 + (\ln 0.30)^2}} = 0.357857131. \tag{2.19}$$

The characteristic polynomial is $s^2 + s/8.1 + (14.2748)(0.93)K_p/8.1$. Matching coefficients gives $\omega_n^2 = AK_p k_s/\tau$ and $2\zeta\omega_n = 1/\tau$, hence

$$K_p < \frac{1}{4\tau A k_s \zeta_{\min}^2} = \frac{1}{4(8.1)(14.2748)(0.93)(0.357857131)^2} = 0.018154432. \tag{2.20}$$

The adopted $K_p = 0.01432$ lies between breakaway and the nominal overshoot limit. The complex branches have fixed decay rate $\sigma = 1/(2\tau) = 0.061728395\ \mathrm{s}^{-1}$; they cannot reach $\sigma = 4/30 = 0.133333\ \mathrm{s}^{-1}$ by changing K_p alone.

2.4 Selected Gain and Pole Calculations

Step 5 – Check the adopted gain. This value is retained from the supplied controller design; the calculations test its performance. Substitution gives

$$q = AK_p k_s = (14.2748)(0.01432)(0.93) = 0.190106076, \tag{2.21}$$

$$G_{cl}(s) = \frac{0.190106076}{8.1s^2 + s + 0.190106076}, \tag{2.22}$$

$$\omega_n = \sqrt{q/\tau} = \sqrt{0.190106076/8.1} = 0.153198845\,\text{rad/s}, \tag{2.23}$$

$$\zeta = \frac{1}{2\sqrt{\tau q}} = \frac{1}{2\sqrt{8.1(0.190106076)}} = 0.402929900, \tag{2.24}$$

$$\omega_d = \sqrt{\omega_n^2 - \sigma^2} = 0.140212308\,\text{rad/s}, \tag{2.25}$$

$$s_{1,2} = \frac{-1 \pm \sqrt{1 - 4(8.1)(0.190106076)}}{2(8.1)} = -0.061728395 \pm j0.140212308. \tag{2.26}$$

The damping predicts $M_p = 100e^{-\pi(0.402929900)/\sqrt{1-0.402929900^2}} = 25.080375\%$. The next angle and magnitude checks verify that these poles lie on the proportional root locus.

2.5 Root-Locus Angle and Magnitude Checks

Step 6 – Verify the pole position. The locus for variable K_p uses $G_0(s) = k_s G_y(s) = (Ak_s/\tau)/[s(s + 1/\tau)]$, where $Ak_s/\tau = 1.638959\ \text{s}^{-2}$. At the selected upper pole $s_d = -0.0617284 + j0.1402123$,

$$\angle s_d = 113.76147^\circ, \qquad \angle(s_d + 1/\tau) = 66.23853^\circ, \tag{2.27}$$

$$\angle G_0(s_d) = -113.76147^\circ - 66.23853^\circ = -180^\circ, \tag{2.28}$$

$$K_p = \frac{1}{|G_0(s_d)|} = \frac{|s_d|\,|s_d + 1/\tau|}{Ak_s/\tau} = \frac{(0.153198845)^2}{1.63895852} = 0.01432. \tag{2.29}$$

Both the negative-feedback angle criterion and magnitude criterion are satisfied. The quadratic roots independently give the same pole pair.

2.6 Nominal Step and Settling Calculation

Step 7 – Evaluate the time response. Substituting into Equation (1.7),

$$y(t) = 5\left[1 - e^{-0.061728395t}\left(\cos(0.140212308t) + 0.44025\sin(0.140212308t)\right)\right]. \tag{2.30}$$

The first peak occurs at $t_p = \pi/\omega_d = 22.405969$ s. Since $\cos\pi = -1$ and $\sin\pi = 0$,

$$y(t_p) = 5[1 + e^{-0.061728395(22.405969)}] = 6.254019\,\text{m}, \tag{2.31}$$

so $M_p = (6.254019 - 5)/5 \times 100 = 25.080375\%$.

Step 8 – Calculate settling. The error envelope satisfies $|y - 5|/5 \le e^{-\sigma t}/\sqrt{1-\zeta^2}$. Solving this bound for 2% gives

$$T_{\text{env}} = -\frac{\ln(0.02\sqrt{1 - 0.4029299^2})}{0.061728395} = 64.80975\,\text{s}. \tag{2.32}$$

Actual settling is found by evaluating the full response, locating the last sample for which $|y - 5| > 0.1$ m, and taking the next sample. A 0.005 s grid gives 54.89 s. This definition agrees with reference-based settling calculations [1].

Step 9 – Check command headroom. Command headroom follows from $u = 2.5 + K_p k_s(5 - y)$: the highest command is 2.566588 V at the step, and the lowest is $2.5 - 0.01432(0.93)(1.254019) = 2.483299$ V at the altitude peak. Neither reaches the 0 or 5 V limit.

2.7 Aerodynamic and Component Uncertainty

With fixed k_T, the uncertain characteristic polynomial is $s^2 + b's + k_T K_p k_s$. The worst scalar case has $b' = 0.9b_0$ and $K_p' = 1.1K_{p,0}$. Applying the required damping ratio gives

$$K_{p,0} < \frac{(0.9b_0)^2}{4k_T k_s \zeta_{\min}^2 (1.1)} = \frac{0.81}{1.1}(0.01815443) = 0.01336826. \tag{2.33}$$

At the supplied gain the worst-case damping is $(0.9/\sqrt{1.1})(0.4029299) = 0.34575$, and Equation (1.6) gives 31.42% overshoot. This explains why passing drag-only uncertainty is insufficient for the joint case.

For independent loop-resistor tolerances, use the U1 coefficients and U2/U3 gains defined in Section 3. Assign each of R_1–R_{10} either 0.9 or 1.1 times its nominal value, then use both drag extremes. There are $2^{10} \times 2 = 2048$ cases. The bias network is held at its nominal Thevenin resistance and zero-error output is recalibrated in each case. Table 2.3 reports this optimistic subset.

Table 2.3: Independent loop-resistor corner predictions.

Quantity	Range / worst value
Feedback gain K_f	0.009072–0.022784
Reference/feedback ratio a_r/a_f	0.672104–1.487864
Absolute DC tracking error	48.7864%
Overshoot about each case's own final altitude	38.9461%

For example, $R_1 = 0.9(100\,\mathrm{k\Omega})$, $R_2 = 1.1(1\,\mathrm{k\Omega})$, $R_3 = 1.1(100\,\mathrm{k\Omega})$ and $R_4 = 0.9(1\,\mathrm{k\Omega})$ give

$$a_f = 0.9/110 = 0.00818182, \quad \frac{a_r}{a_f} = \frac{(1 + 0.00818182)(1.1)/(90 + 1.1)}{0.00818182} = 1.487864. \tag{2.34}$$

Thus a 5 m reference tends to 7.43932 m even after hover recalibration. The type number cannot remove unequal reference and feedback scaling.

2.8 Practical Gain and Ratio Correction

The corner distributions in Figure 2.1 separate dynamic overshoot from DC ratio error.

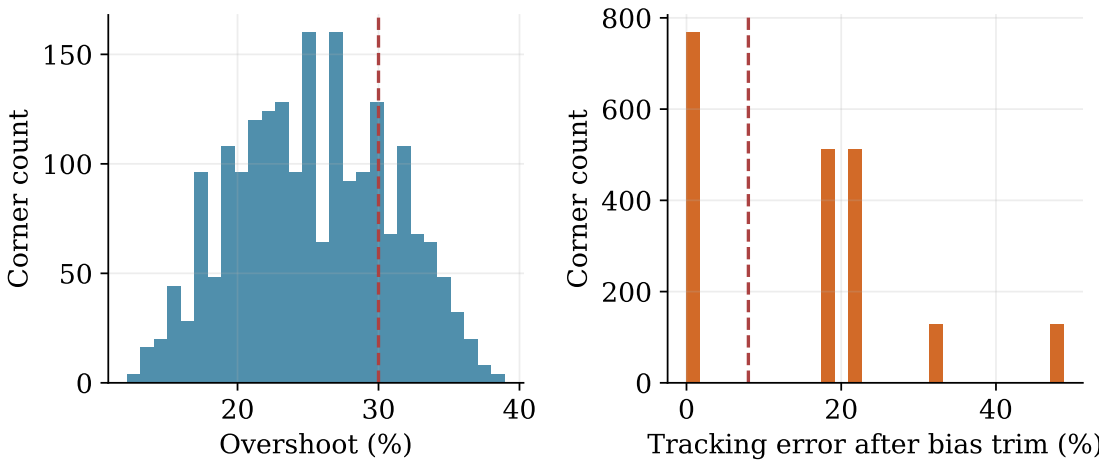


Figure 2.1: Loop-resistor corners. Dashed limits are 30% overshoot and 8% tracking error. The histograms show corner counts, not probabilities. Code: Python; MATLAB. Data: results.

Failure in this subset rejects a guarantee for all components. Bias-network and op-amp variations can add error. After matching the U1 ratios, a practical gain correction is to replace both 1 kΩ resistors with matched 680 Ω values:

$$K_{p,\mathrm{new}} = \frac{680}{100000}(1.47)(0.97427980) = 0.009738901, \tag{2.35}$$

$$\zeta_{\mathrm{worst}} = \frac{0.9b_0}{2\sqrt{k_T(1.1K_{p,\mathrm{new}})k_s}} \approx 0.4193, \tag{2.36}$$

$$M_{p,\mathrm{worst}} \approx 23.44\%, \tag{2.37}$$

$$|\delta u_b|_{\max} = 0.08(5)(0.009738901)(0.93) = 3.623\,\mathrm{mV}. \tag{2.38}$$

This scalar check assumes matched reference and feedback ratios; it is not a guarantee for independent resistor corners. Lower gain increases damping but tightens the hover-bias allowance. The board must be recalibrated and retested after any change.

2.9 Reported Potentiometer Adjustment and Command Calculations

The complete physical three-op-amp controller was connected during the controller test. The student reported adjusting the physical potentiometer until the input was approximately 2.565 V, with the aim of tracking the setpoint with minimal overshoot. No time series or exact measurement instant accompanies that recollection. The retained terminal display independently records a 2.498 V command at 86.10 s.

For the isolated 0-to-5 m reference test, starting at zero altitude,

$$r_v(0^+) = k_s x = 0.93(5) = 4.650\,\mathrm{V}, \tag{2.39}$$

$$e_v(0^+) = r_v - y_v = 4.650 - 0 = 4.650\,\mathrm{V}, \tag{2.40}$$

$$\delta u(0^+) = K_p e_v = 0.01432(4.650) = 0.066588\,\mathrm{V}, \tag{2.41}$$

$$u(0^+) = u_h + \delta u = 2.500000 + 0.066588 = 2.566588\,\mathrm{V}, \tag{2.42}$$

$$u_{\mathrm{reported}} - u(0^+) = 2.565 - 2.566588 = -0.001588\,\mathrm{V}. \tag{2.43}$$

The difference in voltage magnitude is approximately 1.59 mV. This comparison does not establish that the reported observation was measured at the same instant or node as the model command.

For the supplied Simulink startup sequence, define the unit closed-loop response

$$h(t) = 1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right), \quad h(t) = 0 \text{ for } t < 0. \tag{2.44}$$

Superposition gives $y(t) = 2.5h(t) + 2.5h(t - 2.5)$ for a reference of 2.5 m initially and 5 m after 2.5 s. Consequently,

$$u(2.5^+) = 2.5 + 0.01432(0.93)\{5 - 2.5h(2.5)\} = 2.564405\,\mathrm{V}. \tag{2.45}$$

This maximum differs from the reported 2.565 V by approximately 0.595 mV. The original scope's voltage resolution is approximately 0.0124 V/pixel, so it cannot verify either millivolt difference.

The command during a reference error must be distinguished from the zero-error hover bias. If 2.565 V were instead held as a constant zero-error command, its 65 mV excess would predict

$$\delta y = \frac{2.565 - 2.500}{K_p k_s} = \frac{0.065}{0.01432(0.93)} = 4.88076\,\mathrm{m}. \tag{2.46}$$

That is not a negligible hover-bias change. Therefore the reported adjustment is documented as a practical test observation, rather than silently replacing the nominal 2.500 V hover term or the calculated controller gain. Changing the bias alone does not change the damping ratio or remove the proportional controller's settling limitation.

2.10 Experimental Metrics and Uncertainty

The digitised plot uses pixel bounds $x = 602 \ldots 1414$ for 0–120 s and $p_y = 60 \ldots 540$ for 20–0 m. Therefore

$$t = \frac{x - 602}{1414 - 602}(120), \qquad h = \frac{540 - p_y}{540 - 60}(20), \tag{2.47}$$

$$\Delta t = 0.147783\,\text{s/pixel}, \qquad \Delta h = 0.041667\,\text{m/pixel}. \tag{2.48}$$

The extracted peak is 6.229167 m at 22.019704 s. With the displayed reference, terminal height and zero starting altitude,

$$e_f = 5.033 - 4.897 = 0.136\,\text{m}, \tag{2.49}$$

$$\eta_{\text{track}} = 100(1 - 0.136/5.033) = 97.297834\%, \tag{2.50}$$

$$M_{p,\text{ref}} = 100(6.229167 - 5.033)/5.033 = 23.766475\%, \tag{2.51}$$

$$\delta M_{p,\text{ref}} \approx 100(0.041667)/5.033 = 0.82787 \text{ percentage points}. \tag{2.52}$$

This practical overshoot metric uses the commanded reference. If the terminal display is used as a proxy for the actual settled output, the conventional final-output-normalised overshoot is $(6.229167 - 4.897)/4.897 \times 100 \approx 27.20\%$. Both estimates are below 30%, but the screenshot alone does not establish an exact asymptotic output. Table 2.4 compares the analytical, simulated and experimental metrics.

Table 2.4: Comparison of calculation, simulation and retained experiment.

Metric	Analytical	Simulation	Experiment
Reference	5.000 m	5.000 m	5.033 m
Peak	6.25402 m	6.25402 m	$\approx$ 6.229 m
Peak time	22.40597 s	22.405 s	$\approx$ 22.020 s
Reference overshoot	25.0804%	25.0804%	$\approx$ 23.7665%
Tracking accuracy	100% ideal	$\to$ 100%	97.2978% displayed
2% reference settling	64.81 s envelope	54.89 s	Not established

For the 5.033 m reference the 8% band is $5.033(1 \pm 0.08) = 4.63036$–$5.43564$ m. The 2% band is 4.93234–5.13366 m. The terminal height lies inside the first but outside the second. The discrepancy cannot be removed by redefining the settling band around the biased terminal value while claiming reference tracking.

The command-bias calculation uses Equation (1.8): $\delta y = 0.002/(0.01432 \times 0.93) = 0.150177$ m. The illustrative test in Figure 4.1 injects a positive 2 mV bias at 70 s; it is a model prediction rather than a recorded disturbance experiment.

3 Circuit Gain and Bias Loading

With the wiper unloaded, $R_P = P1 \parallel R_{12} = 10 \parallel 1.2 = 1.0714286\ \mathrm{k\Omega}$. The end-to-end current and pot-end voltages are

$$I = \frac{-15}{3.9 + 1.0714286 + 22} = -0.556143\,\mathrm{mA}, \tag{3.1}$$

$$V_A = I(3.9) = -2.168962\,\mathrm{V}, \qquad V_B = I(3.9 + 1.0714286) = -2.764831\,\mathrm{V}, \tag{3.2}$$

$$V_{\mathrm{th}} = V_A + W(V_B - V_A). \tag{3.3}$$

Set the supply sources to zero and inject a 1 A test current at the wiper. The node-conductance equations for A, B and the wiper are

$$\begin{bmatrix} G_{11} + G_{12} + G_t & -G_{12} & -G_t \\ -G_{12} & G_{13} + G_{12} + G_b & -G_b \\ -G_t & -G_b & G_t + G_b \end{bmatrix} \begin{bmatrix} V'_A \\ V'_B \\ V'_w \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix}, \tag{3.4}$$

where $G_i = 1/R_i$, $G_t = 1/(WP1)$ and $G_b = 1/[(1-W)P1]$. The test voltage V'_w gives R_{th}. Solving the hover equation

$$-\frac{R_{10}}{R_9 + R_{\mathrm{th}}(W)} V_{\mathrm{th}}(W) = 2.5 \tag{3.5}$$

gives $W = 0.783080594$, $R_{\mathrm{th}} = 5423.0014\ \Omega$ and $V_{\mathrm{th}} = -2.6355750$ V. The loaded zero-error wiper is -2.500 V.

For U3, define $r = R_{10}/(R_9 + R_{\mathrm{th}}) = 0.94855960$. Since the non-inverting divider gives $V_+ = V_2/2$, virtual short and KCL give

$$u = (1 + r)V_+ - rV_{\mathrm{th}} = \frac{1 + r}{2} V_2 - rV_{\mathrm{th}}, \tag{3.6}$$

$$g_3 = \frac{1 + 0.94855960}{2} = 0.97427980, \tag{3.7}$$

$$K_p = \left(\frac{1}{100}\right)\left(1 + \frac{4.7}{10}\right)(0.97427980) = 0.014321913. \tag{3.8}$$

The ideal buffered result 0.0147 differs by 2.57%, while the selected model value 0.01432 differs from the loaded calculation by only 0.0134%. These are circuit-model calculations; a measured input/output slope of the final board was not supplied.

4 Defined Disturbance Prediction

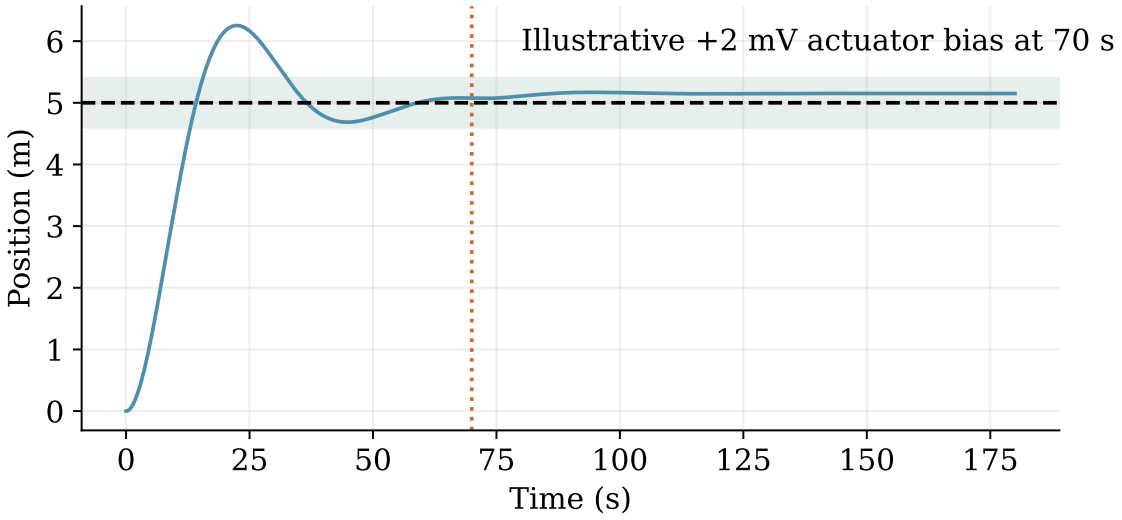


Figure 4.1: Prediction for a +2 mV command bias introduced at 70 s; the shaded band is the 8% reference band. Code: Python; MATLAB. Data: results.

The nominal shift is $0.002/(0.01432 \times 0.93) = 0.150177$ m. This simulation is a prediction for a specified disturbance, not an experimentally measured rejection test.

References

[1] MathWorks, *stepinfo*: official documentation.